

Multiple Choice

1. **Answer:** C

Options: A) It has a fusion crust; B) It contains solidified spherical droplets; C) It is highly processed; D) It has different elemental composition than earth

Note: Correct option: C - It is highly processed

2. **Answer:** C

Options: A) A kilogram of feathers; B) Five pounds of bricks as measured on Earth; C) Five kilograms of feathers; D) A kilogram of bricks

Note: Correct option: C - Five kilograms of feathers

3. **Answer:** A

Options: A) Its rotation is too slow.; B) It has too thick an atmosphere.; C) It is too large.; D) It does not have a metallic core.

Note: Correct option: A - Its rotation is too slow.

4. **Answer:** C

Options: A) viruses; B) bacteria such as E. coli; C) organisms living deep in the oceans around seafloor volcanic vents and in hot springs; D) plankton that use sunlight as an energy source through photosynthesis

Note: Correct option: C - organisms living deep in the oceans around seafloor volcanic vents and in hot springs

5. **Answer:** B

Options: A) Being hit in the head by a bullet.; B) Being hit by a small meteorite.; C) Starvation during global winter caused by a major impact.; D) Driving while intoxicated without wearing seatbelts.

Note: Correct option: B - Being hit by a small meteorite.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '29 Earth days'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Io'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '28000 km/h'.

Long Form Response

11. **Answer:** Their rapid rotation flings the mass near the equator outward.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The two stars will look like a single point of light.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key